

Perceptual Visibility Prediction of Overlaid Content in Optical See-Through Augmented Reality

光学シースルー拡張現実における
重畳コンテンツの知覚的視認性予測

Oishi-Lab M2 Kazuki Hamada

Department of Electrical Engineering and Information Systems

Graduate School of Engineering, The University of Tokyo

2026/05/15

Outline

1. Introduction
2. Related Work
3. Method
4. Pilot Experiment
5. Conclusion & Future Work

Outline

- 1. Introduction** — background, objective
2. Related Work
3. Method
4. Pilot Experiment
5. Conclusion & Future Work

Introduction: Optical See-Through AR

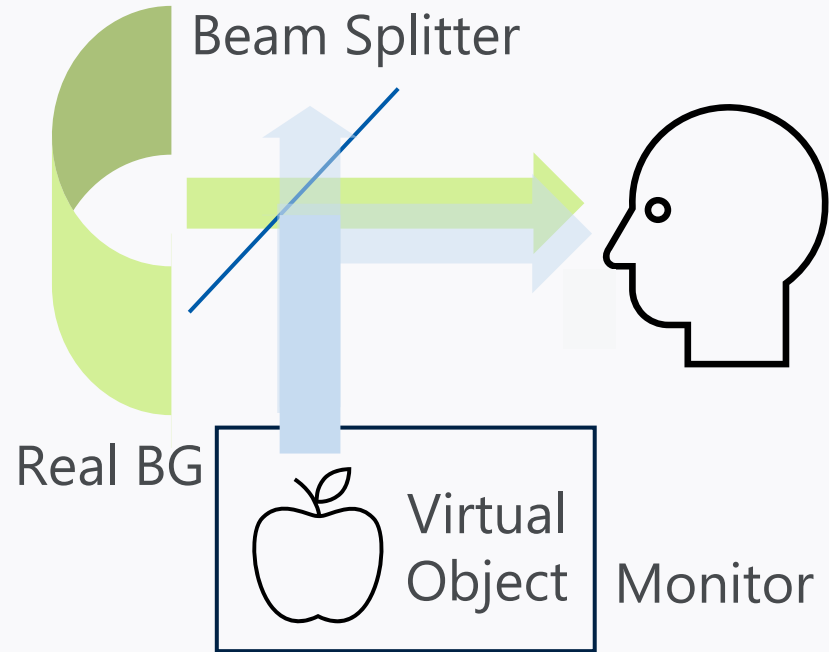
A virtual image is **optically overlaid** on the real-world background

Pros:

Direct perception of real world

Cons:

Additive mixture of FG & BG



Introduction: Visibility Depends on the Background

Visibility of overlayed content varies depending on backgrounds



High visibility



Low visibility
→ **Bad UX, Safety**

→ Need to **predict visibility** from the FG/BG image

Objective

Goal: Build an **image-based visibility predictor** for OST-AR

Three steps

1. **Psychophysical Experiment :** Reveal HVS mechanism
2. **Model Extension:** From 2D to OST-AR
3. **Model Validation:** model's predictions vs human ratings

Outline

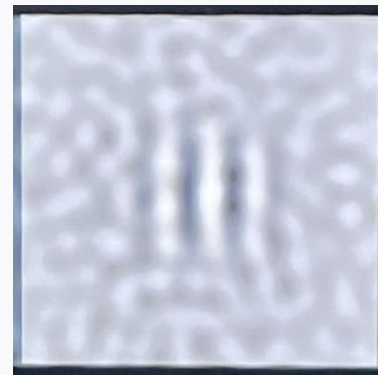
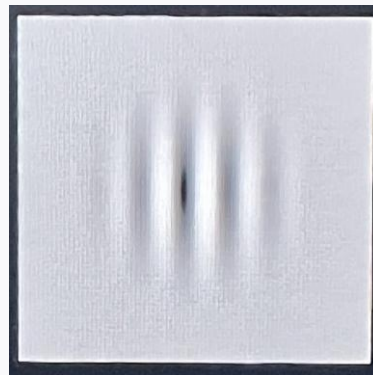
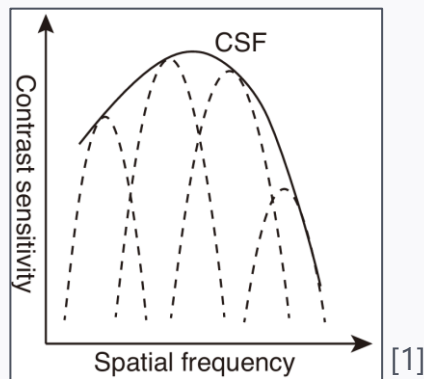
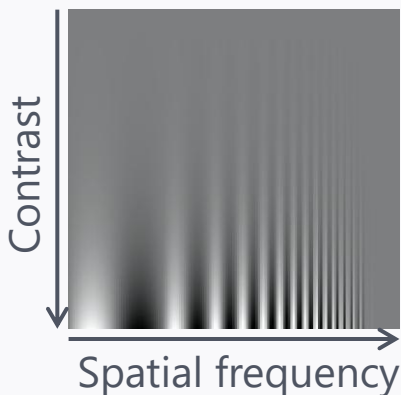
1. Introduction
2. **Related Work** — four topics
3. Method
4. Pilot Experiment
5. Conclusion & Future Work

Related Work (1): Contrast Perception

Visibility ← **Contrast Perception** | Two phenomena addressed

① Frequency dependence (CSF)

② Contrast masking



→ This work mainly focus on ② **contrast masking**

Related Work (2): Single-Factor Models for OST-AR

Existing models each consider **one perceptual factor**

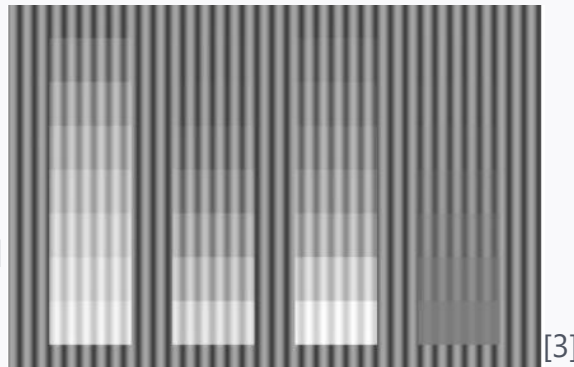
Color



Luminance



Transparency



→ **No unified, multi-factor model exist**

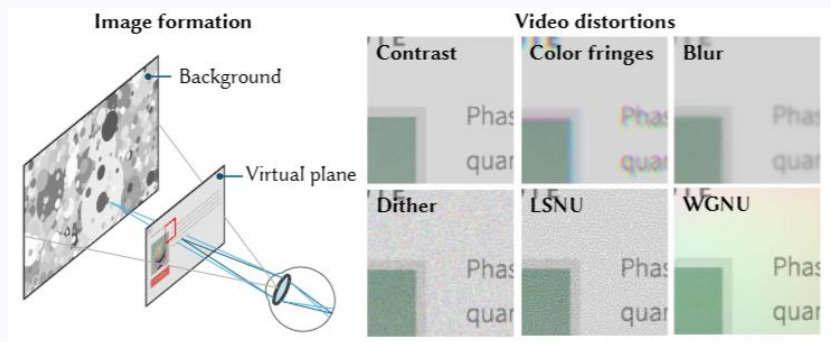
[1] Yoon et al. "Background scene dominant color based visibility enhancement of head-up display," 2017

[2] Liu et al. "How bright should a virtual object be to appear opaque in optical see-through ar?" 2022

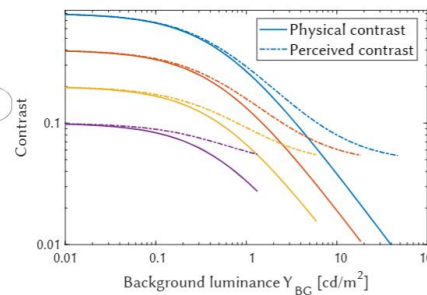
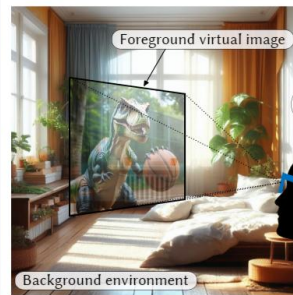
[3] Zhang and Murdoch, "Perceived transparency in optical see-through augmented reality," 2021

Related Work (3): IQA for OST-AR Visibility

IQA(Image Quality Assessment) closely relates to visibility



[1]



[2]

Evaluation of IQA models on OST-AR
→ **limitation of existing model**

OST-AR specific contrast perception
(A potential factor behind the limitation)

→ **OST-AR-specific perception must be modeled.**

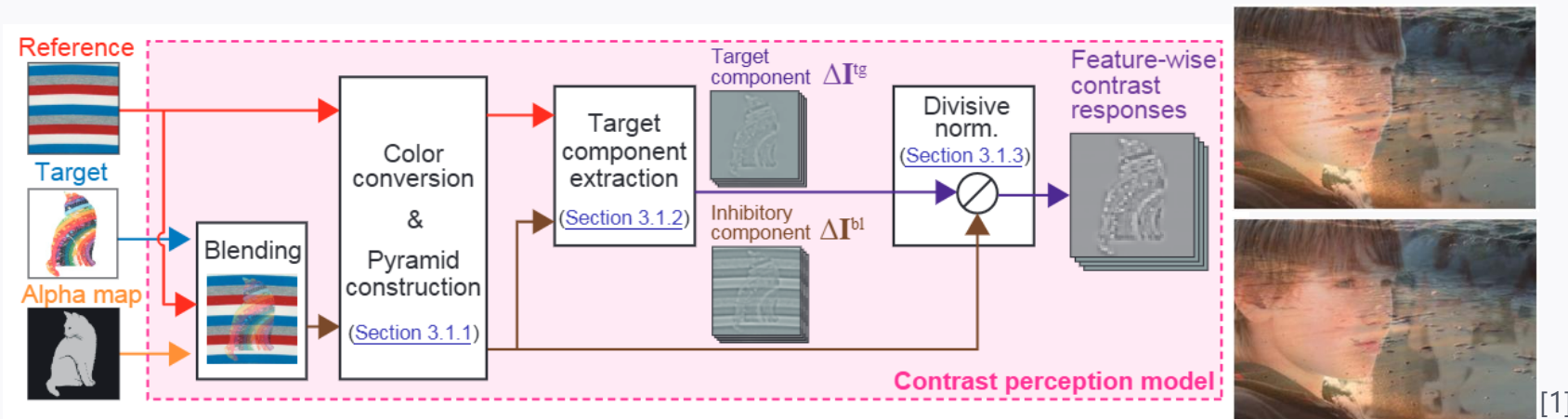
[1] Chapiro, et al. "AR-DAVID: Augmented Reality Display Artifact Video Dataset." 2024

[2] Kim et al. "Supra-threshold Contrast Perception in Augmented Reality." 2025

Related Work (3): Fukiage's Visibility Predictor

Visibility predictor for **α -blended content** — promising for OST-AR

! Considers **multiple perceptual factors** jointly



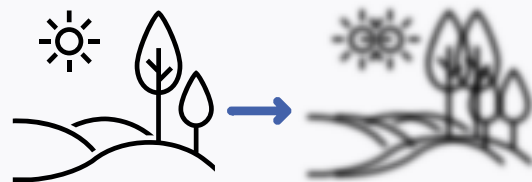
→ But **some gaps** remain when applied to OST-AR

Related Work (4): Fukiage's Visibility Predictor

The model assumes **α -blending** — single 2D display

Four OST-AR specific gaps:

1. **Accommodation:** induces defocus blur
2. **Binocular disparity:** arises with depth difference
3. **BG discounting:** observers partly ignore the real BG
4. **Optical addition:** HDR-range output



→ This work **bridges these gaps**, extending the model to OST-AR

Outline

1. Introduction
2. Related Work
3. **Method** — overview & three steps
4. Pilot Experiment
5. Conclusion & Future Work

Method: Overview

Three steps to extend Fukiage's model for OST-AR

STEP 1 Visibility Evaluation Experiment

— Quantify perceptual effects on visibility

STEP 2 Model Extension

— Extend input processing from results

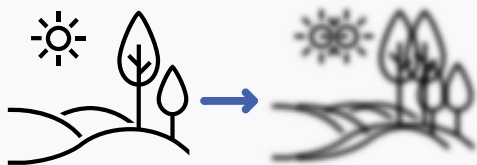
STEP 3 Model Validation

— Compare with human rating

Method (1): Visibility Evaluation Experiment

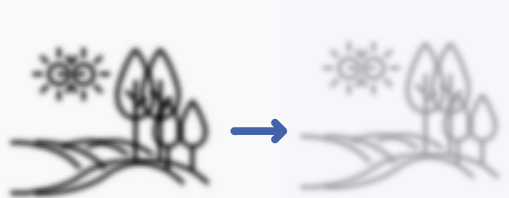
Quantify **how each OST-AR specific factor affect visibility**

BG perception change



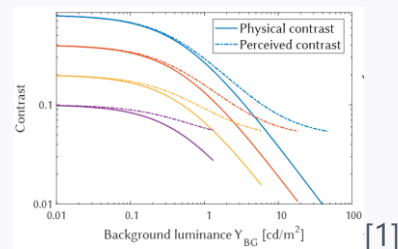
**Defocus
+ Binocular disparity**

BG discounting



**Non-optical
perception change**

Linear summation



**Contrast perception
change**

→ **Decide extensions & fit parameters**

Method (2): Model Extension

Extend model pipeline with **four OST-AR-specific mechanisms**

① **Defocus blur** (Calculate PSF)

$$h(\vec{u}; \Delta D, p) = \begin{cases} 1, & \|\vec{u}\| < p\Delta D/2 \\ 0, & \text{otherwise.} \end{cases}$$

② **Disparity & combination**

Geometric computation
& existing-model selection

③ **Background discounting**

Fit the attenuation process

④ **HDR → SDR** (Tone mapping)

$$L_d(x) = \frac{L(x)}{1 + L(x)} \quad [1]$$

Method (3): Model Validation

Evaluate the model by **comparing with human evaluation**

Psychophysical experiment

Visibility evaluation in OST-AR, using **general texture images**.



Model Evaluation

Quantitatively compare **human scores** with **model predictions**
(e.g., R^2 , SROCC)

Outline

1. Introduction
2. Related Work
3. Method
4. **Pilot Experiment** — design, result, discussion
5. Conclusion & Future Work

Pilot Experiment: Objective

Identify the **parameters & conditions** for the main experiment

Quantify how **defocus blur** & **uncrossed disparity** affect visibility, across:

Stimulus parameters

- Spatial frequency
- FG / BG luminance
- BG contrast

Viewing conditions

- Diopter difference
- Ocularity

Pilot Experiment: Apparatus

Optical components

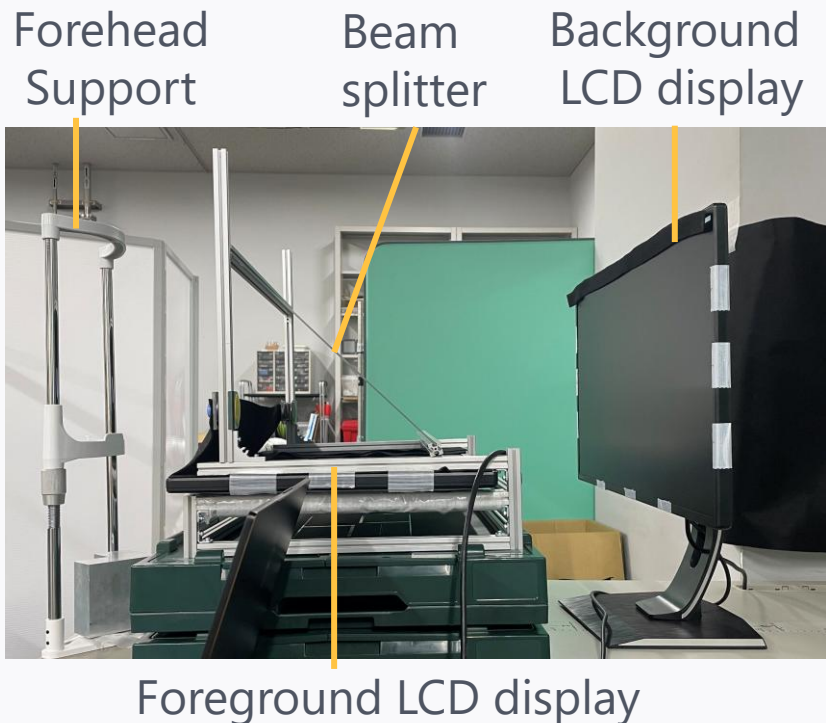
- Beam splitter:
T 20–30%, R 50–60%
- $2 \times 27"$ LCD, 2560×1440

Luminance

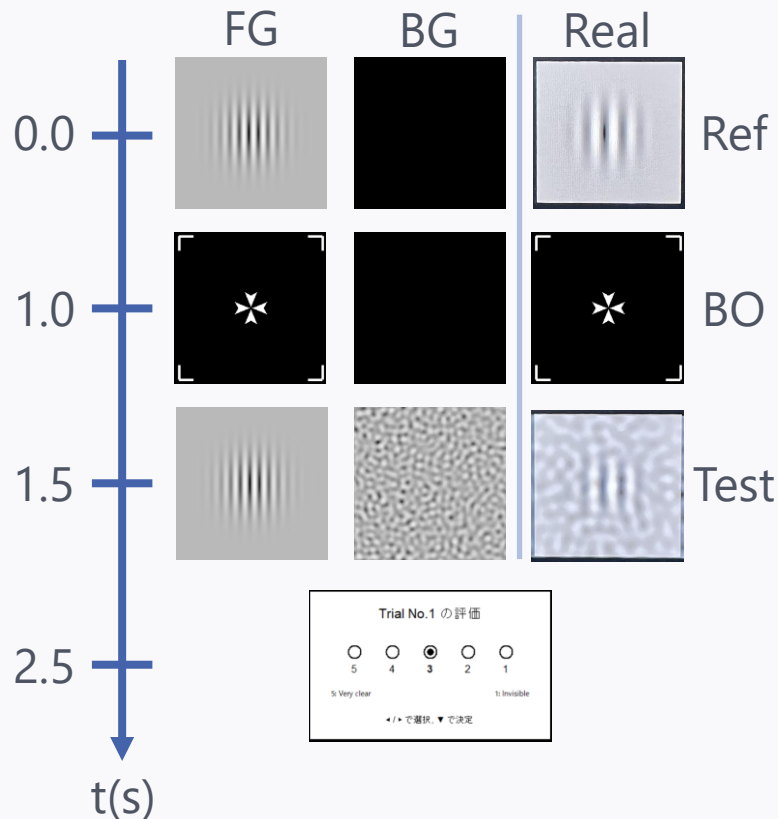
- FG max: 100 cd/m^2
- BG max: 30 cd/m^2

Viewing geometry

- Chin rest, dark room



Pilot Experiment: Procedure



Rating scale

- **1** Invisible — **5** Very clear

Stimulus pairing

- **Reference:** Gabor patch
- **Test:** Gabor + BG noise

Trials

- All combinations × 2 repeats
- Randomized
- N = 1 observer (author)

Pilot Experiment: Stimuli

Specification	Value
Stimuli size	$7.9^\circ \times 7.9^\circ$
Envelope of Gabor patch (σ)	$1^\circ \times 1^\circ$
Spatial frequency (SF, cpd in Gabor patch)	2, 4, 6, 8 cpd
Foreground luminance(Y_{FG})	<u>50 cd/m²</u> or 5 cd/m ²
Background luminance(Y_{BG})	15 cd/m ² or 5 cd/m ²
Foreground contrast(C_{FG})	1.0
Background contrast(C_{BG})	1.0 or 0.0

Pilot Experiment: Conditions

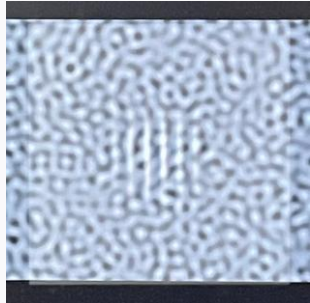
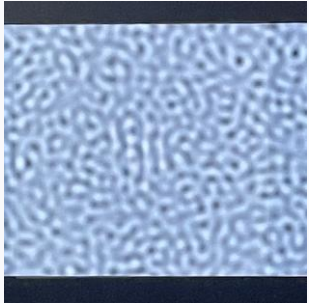
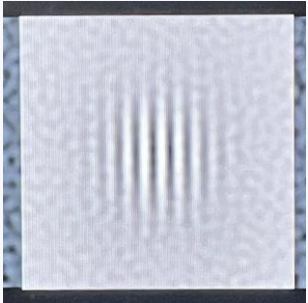
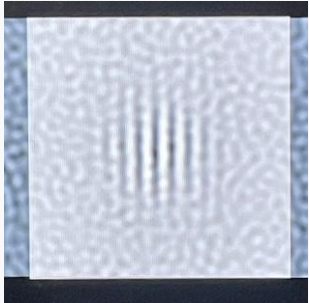
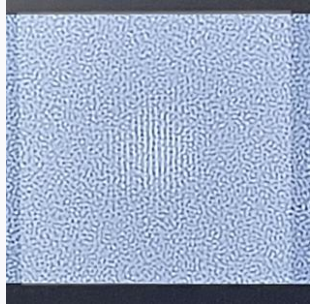
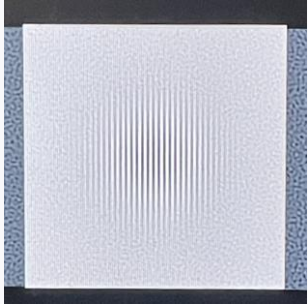
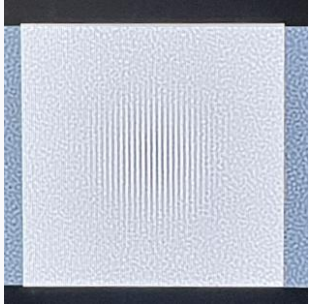
Pair num	d_{FG} [m]	d_{BG} [m]	D_{FG} [D]	D_{BG} [D]	ΔD [D]
1	0.50	0.70	2.00	1.43	0.57
2	0.81	1.50	1.23	0.67	0.57
3	0.50	1.00	2.00	1.00	1.00
4	0.60	1.50	1.67	0.67	1.00

✂ Defocus blur arise **over 0.3 D**

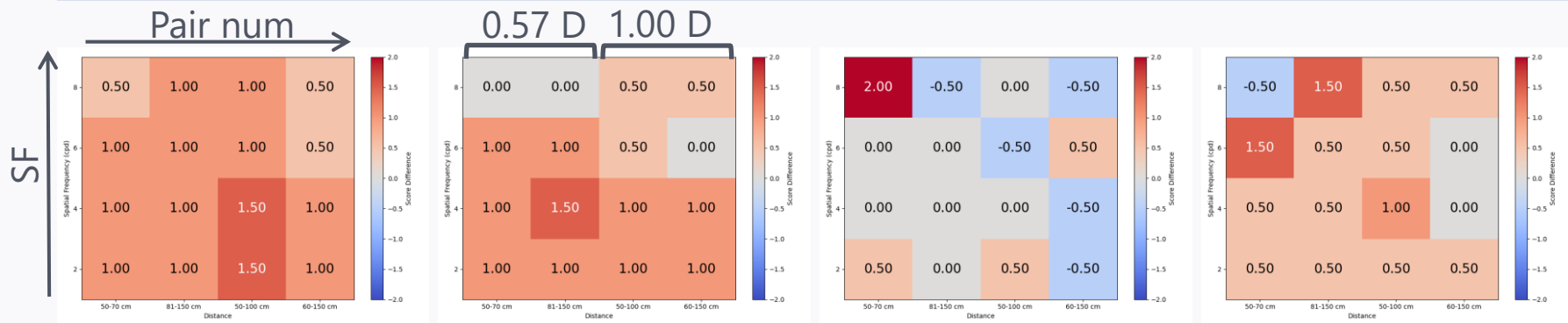
Other conditions

- Monocular (no disparity) vs Binocular (disparity & combination)

Pilot Experiment: Example of Test Stimuli

2 cpd				
6 cpd				
Luminance [cd/m ²]	$\{Y_{FG}, Y_{BG}\}$ $=\{5.0, 5.0\}$	$\{Y_{FG}, Y_{BG}\}$ $=\{5.0, 15.0\}$	$\{Y_{FG}, Y_{BG}\}$ $=\{50.0, 5.0\}$	$\{Y_{FG}, Y_{BG}\}$ $=\{50.0, 15.0\}$

Pilot Experiment: Results — Monocular

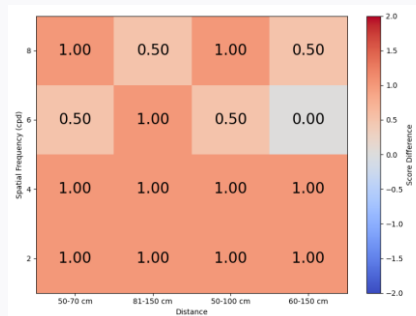


(a) $\{Y_{FG}, Y_{BG}\} = \{5.0, 5.0\} \text{ [cd/m}^2\text{]}$ (b) $\{Y_{FG}, Y_{BG}\} = \{5.0, 15.0\} \text{ [cd/m}^2\text{]}$ (c) $\{Y_{FG}, Y_{BG}\} = \{50.0, 5.0\} \text{ [cd/m}^2\text{]}$ (d) $\{Y_{FG}, Y_{BG}\} = \{50.0, 15.0\} \text{ [cd/m}^2\text{]}$

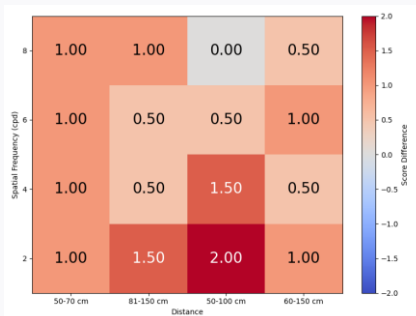
Key observations

- Strongest degradation at (a)(b)
- No consistent ΔD trend
- Low SF (2–4 cpd) more affected
- Some negative scores

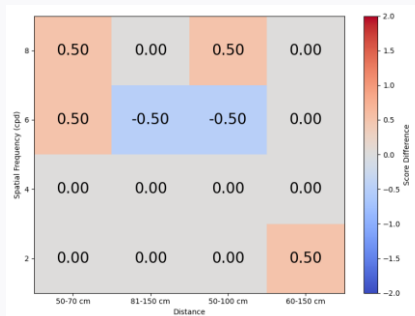
Pilot Experiment: Results — Binocular



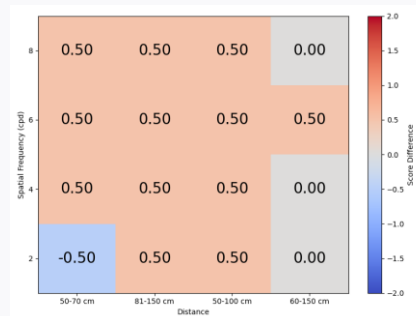
(a) $\{Y_{FG}, Y_{BG}\}$
 $=\{5.0, 5.0\}$ [cd/m²]



(b) $\{Y_{FG}, Y_{BG}\}$
 $=\{5.0, 15.0\}$ [cd/m²]



(c) $\{Y_{FG}, Y_{BG}\}$
 $=\{50.0, 5.0\}$ [cd/m²]

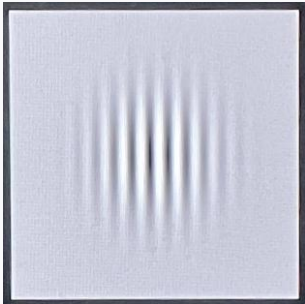
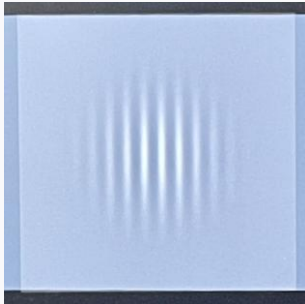
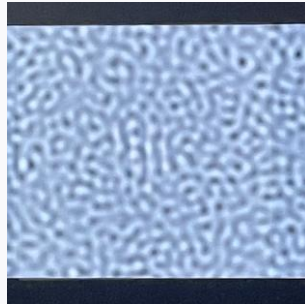
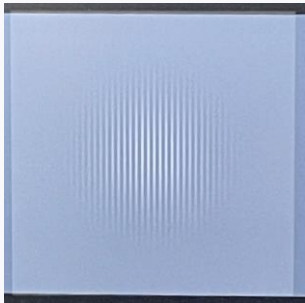
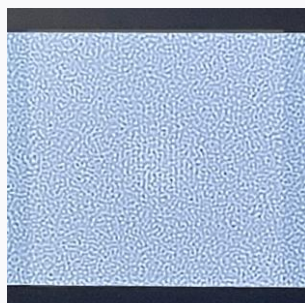


(d) $\{Y_{FG}, Y_{BG}\}$
 $=\{50.0, 15.0\}$ [cd/m²]

Key observations

- Luminance, SF, negative scores patterns same as monocular
- No consistent ΔD trend
- No mono-bino difference

Pilot Experiment: Results — example

	Ref	(b) at $C_{BG}=0.0$	(b) at $C_{BG}=1.0$
2 cpd			
6 cpd			

Pilot Experiment: Discussion

Four redesign directions for the main experiment

① Drop weak masking conditions

Discrete rating lacks resolution

where the true effect is small

② Fix luminance & spatial freq

High Y_{BG} , mid Y_{FG} , and low SF

→reduce other axes

③ Switch to matching method

Use **slower, but more accurate**

method than discrete rating

④ Add single-plane condition

Simulate

linear addition & defocus blur

Outline

1. Introduction
2. Related Work
3. Method
4. Pilot Experiment
5. Conclusion & Future Work

Conclusion & Future Work

Conclusion

- **Goal:** extend Fukiage's visibility predictor to OST-AR through an experiment
- **Pilot experiment:** design study for that experiment
- **Pilot result:** identified the key factors and parameter ranges to target

Future Work

Add pre-experiment
to check redesign



Main experiment
after ethical review



**Model Extension
& Validation**